

SUPRAJA K

Python Developer

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- Over 5 years of experience in designing, developing, and deploying enterprise applications using **Python**, **SQL**, **C**, and **C++**, with strong backend expertise using **Flask** and **Django** frameworks.
- Proficient in **Database management** across various platforms including **MySQL**, **PostgreSQL**, **Oracle**, **MongoDB**, and **SQL Server**, with hands-on experience in writing complex queries and optimizing performance.
- Skilled in **Data engineering** and **analysis** using **PySpark**, with a strong grasp of **Data transformation**, **cleansing**, and **pipeline development** using **AWS Glue** and **ETL best practices**.
- Experienced in **web scraping** (BeautifulSoup, Scrapy), **data visualization** (Matplotlib, Seaborn, Plotly), and foundational **machine learning** using **Scikit-learn** and **TensorFlow**.
- Proficient in deploying and managing cloud-based applications using a wide range of **AWS services** such as **EC2**, **S3**, **RDS**, **VPC**, **Lambda**, **CloudFormation**, **CloudWatch**, and **IAM**, with automation via **AWS CLI** and **CloudFormation templates**.
- Implemented and maintained **RESTful APIs**, integrated **third-party services**, and built **CI/CD pipelines** using **Jenkins**, **AWS CodePipeline**, and **Git** for streamlined development and deployment.
- Familiar with **front-end technologies** (HTML5, CSS3, JavaScript), and experienced in **Agile methodologies** (Scrum, Kanban) to deliver iterative, high-quality software solutions.
- Strong interpersonal and communication skills, with a proven ability to collaborate effectively in cross-functional teams and consistently deliver scalable, maintainable, and secure applications.

PROFESSIONAL EXPERIENCE

STATE FARM

Python Developer

Texas, USA

Nov 2023 – Current

- Built scalable and secure web applications using **Python** and the **Django framework**, utilizing Django ORM for streamlined database operations.
- Developed and integrated **GraphQL** APIs, reducing API response time by 30% and improving front-end performance.
- Managed application data using **MySQL** and **MongoDB**, ensuring high performance, scalability, and data integrity.
- Created responsive, cross-browser-compatible UI components using **HTML5**, **CSS3**, **Bootstrap**, and modern **JavaScript (ES6+)**.
- Developed dynamic front-end features with **React.js**, leveraging **Redux** for state management and **React Router** for client-side routing.
- Architected and implemented microservices using **Express.js**, improving scalability and maintainability by 40%.
- Designed and maintained CI/CD pipelines using **Jenkins** to automate build, test, and deployment processes. Leveraged **Terraform** to implement Infrastructure as Code (IaC) for provisioning and managing AWS infrastructure with improved cost-efficiency and scalability.
- Configured and managed various **AWS services**—including **EC2**, **S3**, **RDS**, **Lambda**, **ELB**, **EBS**, **SQS**, and **SNS**—resulting in a 25% reduction in deployment time and operational costs.
- Implemented **CloudFront** as a content delivery network (CDN) to optimize application load times and reduce latency.

- Worked extensively with XML, including schema definition (XSD) and transformation (XSLT) for effective data processing and exchange.
- Configured **Nginx** as a reverse proxy for load balancing, SSL termination, and serving static assets in production environments.
- Containerized applications using **Docker** to streamline deployment and ensure consistency across development, staging, and production environments.
- Deployed and managed containerized workloads with **Kubernetes**, automating scaling, rollout, and orchestration tasks.
- Utilized **PyCharm** for efficient Python development and debugging, ensuring high code quality and productivity.
- Documented **RESTful** and **GraphQL APIs** using Swagger to provide clear, developer-friendly API references.
- Tracked project progress, issues, and tasks using **JIRA**, enforced coding standards and code quality through reviews using SonarQube.
- Configured **AWS IAM roles** and policies for granular access control and enhanced application security.
- Monitored system metrics and application logs with **AWS CloudWatch**, implementing proactive optimizations for performance and reliability.
- Ensured code robustness and reliability by writing comprehensive unit tests using the **PyUnit** testing framework.

UNITED HEALTH GROUP
Python Developer

Arizona, USA
Aug 2022 – Oct 2023

- Designed and implemented a dedicated **MySQL database server** to support web applications and generate daily performance reports.
- Utilized the **Flask framework** for efficient web application development.
- Used **Python's BeautifulSoup4** library for web scraping and data extraction to build visual data representations. Built dynamic views and templates using Django's view controllers and templating language to create a user-friendly website interface.
- Integrated **Python-based GUI components** to enable interactive front-end features, including selection filters and input controls.
- Developed and tested multiple dashboard features using **Python, Bootstrap, CSS, JavaScript, and jQuery**.
- Created web applications following the **Model-View-Controller (MVC)** architecture using server-side frameworks like Django and Flask.
- Wrote Python scripts to automate file transfers from Dropbox to **Amazon Web Services (AWS)**. Worked on both front-end and back-end components using Python and the Django web framework.
- Built consumer-facing features and applications using **Django** and **Python** within a test-driven development (TDD) and pair programming environment.
- Integrated a continuous integration (CI) system with a **Git** version control repository to automate builds triggered by code check-ins.
- Designed an interactive communication page using .NET with Microsoft Visual Basic 2014 and SQL development tools.
- Worked on writing **Ansible** playbooks using Python and SSH to manage AWS node configurations and deployed them on development servers using automated Ansible scripts.
- Consumed third-party APIs and developed **RESTful APIs** using **Django REST Framework** in conjunction with **AngularJS**.

- Participated in all phases of the **Software Development Life Cycle (SDLC)** including analysis, design, implementation, and testing, following Agile development practices.
- Utilized **AWS CloudWatch** for monitoring applications and storing log data.
- Collaborated with end-users and clients to gather and analyze system requirements.
- Built the application using **Python** with Spring Inversion of Control (IoC) and the **Django** framework, incorporating Spring Security for access control.
- Developed front-end features to consume **RESTful web services** using **JavaScript** and **AngularJS** for handling JSON and XML data.
- Created and maintained unit tests using **Python's unit test** module and addressed bugs and defects with tools like Firebug.
- Updated **Apache Web Server** TLS/SSL configurations provided staff training on best practices, and automated configuration updates using Ansible.
- Enhanced the existing user interface using **HTML, CSS, jQuery, and Bootstrap**.
- Worked extensively with **HTML5, CSS3, JSON, JavaScript, and ES6** to design dynamic and responsive UI components.
- Leveraged **JIRA** for task management, issue tracking, reporting, and auditing.
- Implemented business logic and handled data transformation for XML and HTML using **Python 2.7** and **Django**.
- Performed database performance tuning and optimized stored procedures and SQL queries.
- Applied object-oriented programming principles to improve the functionality and maintainability of product management systems.

TECHNICAL SKILLS

Languages	Python, Java, C, C++
Frameworks	PySpark, Django, Flask, Angular, Docker and Spring MVC
Web Technologies	HTML5, CSS3, Bootstrap, JavaScript, jQuery
Libraries	PyVISA, PySpark, NumPy, Pandas, Matplotlib, Scikit-learn, OpenCV, Scrapy, TensorFlow, PyTorch, SciPy
Java/J2EE Technologies	Core Java, JSON, Java Beans, JDBC, Rest API
Application Servers	Apache2, JBoss, WebLogic
Database	MySQL, MongoDB, Oracle
IDE's	Visual Studio, PyCharm, Eclipse
Operating Systems	Windows, Linux, Unix
Version Control	SVN, GIT

ACADEMIC PROJECTS

AWS Cloud Application – Image Recognition as a Service

Spring 2023

- Developed a scalable image categorization web application using AWS services under the IaaS model, leveraging a multi-tier architecture with a separated web and auto-scaling application tier to efficiently manage varying traffic loads.
- Implemented dynamic scaling policies using AWS Auto Scaling and SQS to optimize resource utilization, ensure high performance under load, and enable cost-effective management of incoming image processing requests.
- Streamlined image storage and retrieval by integrating AWS S3 with versioning and lifecycle policies, ensuring data durability, low-latency access, and robust security for large image datasets.

Full Stack E-Commerce Web Application

Fall 2022

- Developed a dynamic subscription management system with an intuitive user interface using React, HTML, and CSS, enabling users to browse products, subscribe, and manage their plans seamlessly.
- Designed and implemented secure backend services using Node.js and MySQL, with RESTful APIs and JWT-based authentication to ensure secure and reliable user access.
- Enhanced user experience by integrating personalized product recommendations based on user history and offering customizable subscription options tailored to individual preferences.

Audio-Based Confidence Analyzer with Machine Learning

Spring 2022

- Designed and built a machine learning model using Python and Scikit-learn to classify confidence levels in human speech, based on a custom dataset of 500 labeled audio samples from interviews and presentations.
- Collected, labeled, and preprocessed speech data, assigning confidence scores (low to high), and extracted relevant audio features to train and fine-tune the classification model.
- Evaluated model performance on diverse datasets, analyzing bias, variance, and generalization across different speaking contexts to improve robustness and accuracy.

Event Registration Application

Fall 2021

- Built an image categorization web application on AWS using the IaaS model, with a multi-tier architecture that separated the web and auto-scaling application tiers for efficient traffic handling and performance optimization.
- Implemented dynamic scaling and asynchronous request handling using AWS Auto Scaling and SQS, enabling cost-effective resource management and smooth performance during high traffic loads.
- Managed large-scale image data using AWS S3 with versioning and lifecycle policies, ensuring secure, highly available, and low-latency access to image datasets.

EDUCATION

ARIZONA STATE UNIVERSITY

Master of Science in Information Technology

GPA: 4.0/4.0

Tempe, Arizona

May 2023